Elias Dunham

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EDUCATION

Northeastern University, College of Engineering **Boston, MA**
Candidate for Bachelor of Science in Mechanical Engineering and Physics *May 2028*
GPA: 3.8
Coursework: Fluid Mechanics, Thermodynamics, Mechanics of Materials, Materials Science, Classical Dynamics
Honors & Extracurriculars: Dean's List, Commended National Merit Scholar, FBLA State Finalist, American Society of Mechanical Engineers, Northeastern Composing Club, Delta Tau Delta

Northeastern University, Dialogue of Civilizations **Panama City, Panama**
International Applications of Fluid Mechanics *July 2026*
Studied fluid mechanics via case studies at the Panama Canal, a Penonomé wind farm, and Chiriquí hydroelectric facilities

COMPUTER & MACHINING SKILLS

CAD: SolidWorks, Autodesk Fusion, AutoCAD
Programming: C++, JavaScript, Python, MATLAB, Arduino
Manufacturing: 3D Printing (SLA, FDM, PμSL), CNC (3- & 5-axis), Manual Injection & Vacuum Compression Molding, Rotary Axis Laser Cutting
Testing & Instrumentation: SEM, Mechanical Testing (Instron, ADMET), DMM Data Logging, Thermocouple Instrumentation
Electronics: PID Control Systems, Oscilloscopes, Soldering, Circuit Engraving (Laser Ablation)

PROFESSIONAL EXPERIENCE

MIT Lab for Translational Engineering **Cambridge, MA**
Technical Associate I *August 2026 - Present*
- Conduct materials research for a US Navy-funded program on lightweight, high-strength materials for ballistic protection, drone efficiency, and wetsuit insulation, characterizing candidates through high-pressure cycling, hardness testing, and scanning electron microscopy (SEM)

Mechanical Engineering Co-op *January - June 2026*
- Led mechanical design and fabrication across 6 concurrent device programs, taking each from CAD through prototyping and in-vivo validation
- Fabricated device hardware on SLA and FDM printers, a 5-axis mill, and rotary-axis laser cutter, producing over 50 prototypes to support ongoing device programs
- Established repeatable fabrication and test protocols, training 4 lab members to operate independently

FLX Solutions **Bethlehem, PA**
Mechanical Engineering Intern *June - August 2025*
- Replicated the company's embedded-software hardware testbed, authoring assembly instructions, annotated wiring diagrams, and test procedures that let an engineer with no prior exposure rebuild it
- Trained a Microsoft TrOCR model in Python to convert handwritten notes into searchable database records, digitizing over 1,000 records at 87% accuracy

PROJECTS

Wetsuit Thermal Research | US Navy, MIT Lab for Translational Engineering *March 2026 - Present*
- Engineer and operate a high-pressure testing apparatus with a PID-controlled heated surface (37.2°C) at pressures up to 104 fsw, validating performance in submerged and dry conditions with epoxy-potted heater and thermocouple feedthroughs
- Synthesize and evaluate 24 candidate formulations, characterizing structural integrity through high-pressure cycling

Electro-Thermal Actuation for GI Drug Delivery | MIT Lab for Translational Engineering *January - June 2026*
- Characterized Nitinol activation temperatures and nichrome heater resistance and power across lengths and voltages, validating results against Joule-heating theory
- Quantified the operational power envelope of each actuation path against pill-scale size and energy budgets, ruling out a capacitor-pulse scheme and narrowing the design space

In-Situ Mixing Syringe | MIT Lab for Translational Engineering *February - June 2026*
- Designed a dual-barrel syringe in Fusion 360 isolating two components until they mix at the tip, validated for injection through a 23G needle
- Diagnosed leakage from stress concentration at the gasket sealing interface, redistributing sealing load across 11 SLA-printed iterations to eliminate leakage across 8 consecutive trials

Stem-Cell Delivery Device | MIT Lab for Translational Engineering *February - June 2026*
- Redesigned the catheter's central channel to reduce taper and flow impedance, then manufactured the 42 devices and 336 printed and machined parts used for in-vivo experiments

Vagus Nerve Stimulation PCB Housing | MIT Lab for Translational Engineering *February - April 2026*
- Designed and fabricated a clear SLA housing contoured to the animal's back, converging on assembly tolerances across 15 iterations to keep wire ports and snap-fit lid sealed against fluid ingress throughout 4 in-vivo trials